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Effectiveness of cow's milk against zucchini squash powdery mildew (*Sphaerotheca fuliginea*) in greenhouse conditions

Wagner Bettiol*

Brenno Domingues Astiarraga and Alfredo José Barreto Luiz Embrapa Environment, CP 69, 13.820-000 Jaguariúna, SP, Brazil

Abstract

Efficacy of fresh cow milk was tested in five greenhouse experiments against powdery mildew (*Sphaerotheca fuliginea*) on zucchini squash (*Cucurbita pepo*). Plants were sprayed with milk at 5, 10, 20, 30, 40, and 50%, either once or twice a week. Additional treatments were fungicides (fenarimol 0.1 ml/l or benomyl 0.1 g/l) applied once a week and water as a control treatment. Severity of the powdery mildew was visually evaluated on individual leaves at weekly intervals and scored as percentage of leaf area infected for infected leaves. A negative correlation was found between the infected leaf area per infected leaf and milk concentration sprayed on plants for the five experiments. High concentrations of milk were more effective than the conventional fungicides tested. This study demonstrated that milk is an effective alternative for the control of powdery mildew in organic agriculture. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: Powdery mildew; Zucchini squash; Milk; Alternative control; Organic agriculture

1. Introduction

Cucurbit powdery mildew, caused by *Sphaerotheca fuliginea* (Schlecht.) Pollacci, is a serious disease on cucurbits grown worldwide. Powdery mildew occurs on leaves, stems and fruits. Control methods currently available under commercial conditions include the use of repeated applications of elemental sulphur (Kimati et al., 1980) and other fungicides (Kimati et al., 1997). The constant use of fungicides, however, can result in the environmental contamination and selection of resistant populations of *S. fuliginea* (McGrath, 1996; McGrath et al., 1996). For these reasons, alternative control measures are warranted.

Several alternatives to conventional fungicides have been evaluated for cucurbit powdery mildew. Reuveni et al. (1995) and Garibaldi et al. (1994) verified that powdery mildew was controlled significantly by a single spray of aqueous solutions containing various phosphates and potassium salts. These authors concluded that phosphates and potassium are appropriate foliar fertilizers, with a potential beneficial influence on disease

control. Pasini et al. (1997) verified that JMS Stylet oil, canola oil, Synertrol, and neem extract provided satisfactory control of powdery mildew on rose. Marco et al. (1994) described the suppression of powdery mildew on squash by applications of whitewash, clay, and antitranspirant materials. Sodium and potassium bicarbonates combined with Sunspray ultra-fine spray oil, both at 0.5%, were effective in the control of powdery mildew (Ziv and Zitter, 1992). *Ampelomyces quisqualis* has been described as a biological control agent for cucurbit powdery mildew (Falk et al., 1995; McGrath and Shiskoff, 1996; Pasini et al., 1997).

The objective of this work was to find possible antifungal compounds that would provide effective disease control under practical conditions, while also minimizing environmental impacts. We have demonstrated that the residue of glutamic fermentation of molasses and a product of milk fermentation released by *Lactobacillus* were effective for the control of powdery mildew on zucchini squash (Bettiol, 1996; Astiarraga and Bettiol, 1997; Bettiol and Astiarraga, 1998).

This paper reports on the control of powdery mildew on zucchini squash with milk under greenhouse conditions. This study was conducted to obtain an alternative for controlling powdery mildew in organic agriculture. Preliminary information has been reported recently (Bettiol and Astiarraga, 1997).

* Tel.: + 55-19-867-8700; fax: + 55-19-867-8740.

E-mail address: bettiol@cnpma.embrapa.br (W. Bettiol)

2. Materials and methods

Zucchini squash plants (*Cucurbita pepo*, cv. Caserta – CAC) were grown in 2.0 l aluminium pots containing oxisol and composted cattle manure (3 : 1) amended with 10 g of NPK fertilizer (4-14-8). Plants were kept in a greenhouse without inoculum of *Sphaerotheca fuliginea* until they reached the developmental stage of four expanded leaves. At this stage, plants were transferred to a greenhouse with high *S. fuliginea* inoculum potential. They were sprayed to runoff with either fresh cow milk [5, 10, 20, 30, 40, and 50% (per volume in water)] once (first, second, and third experiment) or twice (fourth and fifth experiment) a week, or fungicide (fenarimol 0.1 ml/l or benomyl 0.1 g/l) once a week. Check plants were treated with water only. Applications were performed with a compressed-air paint sprayer at 10 lb/pol². The first application was made immediately after the plants were transferred to the greenhouse with high *S. fuliginea* inoculum. Experiments were set up in a randomized design with five replicates per treatment. Each replication consisted of one pot containing one plant. In order to insure randomization, the pots' positions were rerandomized once a week. The temperatures in the greenhouse during the experiments varied between 20 and 32°C. Zucchini squash plants infested with powdery mildew were placed under the ventilating fans of the greenhouse to distribute spores.

The severity of powdery mildew infestation was visually evaluated on individual leaves, and scored as percentage of area infected at weekly intervals (Garibaldi et al., 1994; McGrath and Shiskoff, 1996).

Statistical analyses were done using GLM and NLIN procedures of SAS (SAS, 1993). The analysis of variance (ANOVA) for all trials and the contrasts of fungicide with the control and each of the milk dosages was done using the *F* test to evaluate significance. A nonlinear regression model was fit to the data with infected leaf area as a function of milk dosages, using the control treatment (water) as the zero milk dosage (0%). The regression model was $Y = \beta_0 - \{\beta_1 \times [1 - \exp(-\beta_2 \times X)]\}$, where *Y* = infected leaf area; *X* = milk dosage; β_0 = mean infected leaf area in the absence of disease control; β_1 = maximum reduction of the infected leaf area obtained by milk treatment; and β_2 = parameter associated with the speed in which the maximum is reached (Souza, 1998). The parameter estimates were used to calculate EC₉₀ values (the milk concentration that reduced infected leaf area by 90%).

3. Results and discussion

Powdery mildew severity was very high in all the experiments on zucchini squash plants receiving the control treatment of water applied once or twice a week.

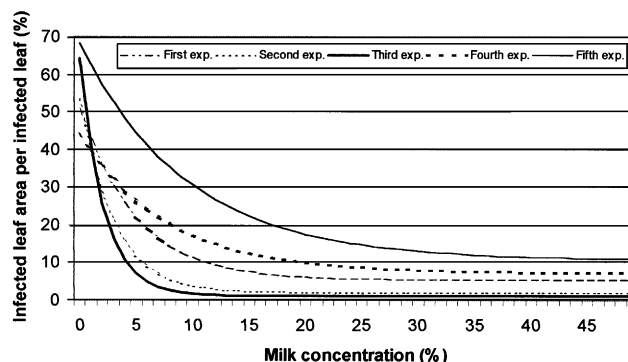


Fig. 1. Relationship of powdery mildew (*Sphaerotheca fuliginea*) severity on leaves (%) of zucchini squash to concentration of milk sprayed twice a week at 22 d after the first spray ($R^2 = 0.94$; 0.96 and 0.94, for the first, second and third experiment, respectively), and once a week at 24 d after the first spray ($R^2 = 0.89$ and 0.96, for the fourth and fifth experiment, respectively).

Infected leaf area per infected leaf exceeded 50% by 22 d after the first spray. A negative correlation was found between the infected leaf area per infected leaf and milk concentration sprayed on plants for the five experiments (Fig. 1).

Milk applied twice a week at concentrations of 10% and higher controlled powdery mildew on zucchini squash at least as effectively as conventional fungicides (Table 1). Several concentrations of milk were more effective than the fungicides fenarimol and benomyl. When the applications were done once a week, milk concentrations of 20 and 50% were required to achieve the same level of disease control as fungicides in the fourth and fifth experiments, respectively (Table 2). Mold grew on the adaxial side of the leaves treated with milk at concentrations of 30% and higher, but plants did not appear to be injured.

The goodness of fit of the regression model was measured by the squared correlation coefficient (R^2) between the observed and predicted values. Estimated parameters and R^2 for each trial are in Table 3. EC₉₀ values for experiments 1–4 indicate that milk dosages ranging from 3.8 to 19.7% would reduce disease severity by 90%. EC₈₅ was calculated with data from the fifth experiment because 90% reduction was not reached with the tested dosages.

Cow milk may have more than one mode of action in controlling zucchini squash powdery mildew. Fresh milk may have a direct effect against *S. fuliginea* due to its germicidal properties (Salle, 1954). Milk contains several salts and amino-acids (Martins Filho, 1987). These substances have been shown to be effective in controlling powdery mildew and other diseases (Reuveni et al., 1993, 1995; Mucharromah and Kuc, 1991; Titone et al., 1997; Pasini et al., 1997). Several authors have shown that sodium bicarbonate, oxalate, dibasic or tribasic potassium phosphate, and other salts and amino-acids

Table 1
Efficacy of milk sprayed twice a week to control zucchini squash powdery mildew (*Sphaerotheca fuliginea*)

Treatment	First experiment		Second experiment		Third experiment	
	% infected leaf area per infected leaf		% infected leaf area per infected leaf		% infected leaf area per infected leaf	
	Days after the first spray		Days after the first spray		Days after the first spray	
	22 d	29 d	15 d	22 d	15 d	22 d
Water	[>] 50.72 ^a	[>] 56.94	[>] 32.46	[>] 53.29	[>] 39.99	[>] 64.23
Milk 5%	[>] 21.71 (57)	[>] 17.47 (62)	[>] 10.99 (66)	[=] 10.51 (80)	[<] 6.32 (84)	[<] 7.06 (79)
Milk 10%	[=] 11.52 (77)	[=] 9.99 (82)	[=] 7.97 (75)	[=] 7.10 (87)	[<] 2.98 (93)	[<] 1.92 (97)
Milk 20%	[<] 7.47 (85)	[=] 5.45 (90)	[=] 2.84 (91)	[<] 2.75 (95)	[<] 1.23 (97)	[<] 1.30 (98)
Milk 30%	[<] 6.63 (87)	[=] 5.06 (91)	[=] 1.24 (96)	[<] 1.84 (97)	[<] 1.28 (97)	[<] 1.35 (98)
Milk 40%	[<] 4.61 (91)	[=] 3.53 (94)	[=] 0.86 (97)	[<] 0.82 (98)	[<] 0.50 (99)	[<] 0.75 (99)
Milk 50%	[<] 4.23 (92)	[=] 3.79 (93)	[=] 0.50 (98)	[<] 0.50 (99)	[<] 0.65 (98)	[<] 1.00 (98)
Fungicide	15.66 (69)	8.80 (85)	2.95 (91)	7.53 (86)	15.13 (62)	15.58 (75)
<i>F</i> (ANOVA)	65.92 ^b	89.18 ^b	30.37 ^b	133.25 ^b	23.9 ^b	61.29 ^b
CV (%)	27.9	30.7	58.4	31.5	67.3	50.4

^aSymbols in square brackets indicate if the infected leaf area is greater [>], equal [=], or smaller [<] in contrast with the fungicide by the *F* test (*P*=0.05). The numbers in parenthesis are the percentage of disease control in relation to the control treatment. The first and the third experiments received fenarimol (0.1 ml/l) and the second experiment benomyl (0.1 g/l). The fungicides were sprayed once a week.

^bIndicates significance at *P*=0.01.

Table 2
Efficacy of milk sprayed once a week to control zucchini squash powdery mildew (*Sphaerotheca fuliginea*)

Treatment	Fourth experiment		Fifth experiment	
	% infected leaf area per infected leaf		% infected leaf area per infected leaf	
	Days after the first spray		Days after the first spray	
	24 d	31 d	17 d	24 d
Water	[>] 44.20 ^a	[>] 56.28	[>] 53.65	[>] 68.55
Milk 5%	[>] 24.77 (39)	[>] 35.31 (37)	[>] 33.41 (38)	[>] 44.10 (36)
Milk 10%	[>] 17.84 (60)	[=] 14.31 (75)	[>] 21.27 (60)	[>] 29.86 (56)
Milk 20%	[=] 10.56 (76)	[=] 9.18 (84)	[>] 14.70 (73)	[>] 18.93 (72)
Milk 30%	[=] 10.23 (77)	[=] 8.31 (85)	[>] 14.55 (73)	[>] 13.07 (81)
Milk 40%	[=] 7.24 (84)	[=] 8.10 (85)	[>] 10.01 (81)	[>] 11.87 (83)
Milk 50%	[=] 4.59 (90)	[=] 4.82 (91)	[=] 8.75 (84)	[=] 9.58 (86)
Fenarimol 0.1 ml/l	5.74 (87)	7.88 (86)	3.72 (93)	5.09 (93)
<i>F</i> (ANOVA)	59.90 ^b	26.54 ^b	67.77 ^b	125.08 ^b
CV (%)	27.0	48.1	22.2	17.2

^aSymbols in square brackets indicate if the infected leaf area is greater [>], equal [=], or smaller [<] in contrast with the fungicide by the *F* test (*P*=0.05). The numbers in parenthesis are the percentage of disease control in relation to control treatment.

^bIndicates significance at *P*=0.01.

have been efficient in the induction of systemic resistance (Reuveni et al., 1993, 1995; Mucharromah and Kuc, 1991; Titone et al., 1998; Pasini et al., 1997; Van Andel, 1966). Therefore milk may also indirectly affect *S. fuliginea* by inducing systemic resistance.

Milk is not a potential environmental or food contaminant, consequently it can be used in organic agriculture. Several organic growers have been spraying 5% cow milk once a week to control powdery mildew on zucchini squash and cucumber.

Table 3

Estimated parameters, squared correlation coefficients and EC₉₀ values for nonlinear regression models fitted to data from each of the five experiments

Milk applications	Trial	Days after first spray	Non linear regression parameters and coefficients				
			β_0^a	β_1	β_2	R ²	EC ₉₀
Twice a week	First	22	50.63847	45.18634	0.20096	0.94	11.4
		29	56.81017	52.19949	0.26584	0.95	8.5
	Second	15	32.08936	30.71073	0.19662	0.85	6.5
		22	53.16968	51.12401	0.33233	0.96	5.6
	Third	15	39.97214	38.87656	0.38995	0.86	3.8
		22	64.22808	63.08247	0.47188	0.94	4.2
Once a week	Fourth	24	43.66734	36.55757	0.12877	0.89	19.7
		31	57.22086	50.90511	0.14216	0.80	18.5
	Fifth	17	53.55440	42.88504	0.12899	0.92	17.8 ^b
		24	68.24291	57.69084	0.10596	0.96	24.2 ^b

^a β_0 = mean infected leaf area in the absence of disease control; β_1 = maximum reduction of infected leaf area obtained by milk treatment; and β_2 = parameter associated with the speed with which the maximum is reached.

^bCorresponds to EC₈₅ values and not to EC₉₀.

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